Supplementary Material of "Hypergraph-Based Fuzzy Cognitive Maps for Functional Connectivity Analysis on fMRI Data"

Yingzhi Teng, Kai Wu, *Member, IEEE*, Jing Liu, *Senior Member, IEEE*, Yifan Li

Appendix A

We need to preprocess the original functional images and extract the fMRI time series of ROIs. fMRI data are preprocessed using Data Processing Assistant for Resting-State fMRI package [1] (DPARSF, http://rfmri.org/DPARSF). The steps are as given follows: 1) Slice timing: the initial 10 time points are removed, and the slice-timing correction is performed; 2) Realign: the subjects with head motion exceeding 2.5 mm and 2.5 degrees are excluded; 3) Co-registered: the functional images are co-registered to the T1-weighted structural images, which are further segmented into gray matter, white matter, and cerebrospinal fluid by the segmentation method in SPM (http://www.fil.ion.ucl.ac.uk/spm); 4) Normalized: the functional images are normalized into the Montreal Neurological Institute (MNI) space by the Diffeomorphic Anatomical Registration Through Exponentiated Lie algebra (DARTEL); 5) Smooth: to reduce the uncertainty of the co-registered, the functional images are smoothed by a 4mm full with at half maximum (FWHM) Gaussian kernel; 6) Filter: functional time series are band-pass filtered at 0.01-0.1 Hz.

According to the Automated Anatomical Labeling template (AAL) [2], a total of 116 non-overlapping ROIs are mapped to cover the whole brain. Then, we extract the fMRI time series of each ROI.

Appendix B

PC: Pearson's correlation (PC) is widely used in neuroimaging [3]–[5]. PC is used to construct FCNs, and SVM is used for classification.

Lasso-HCC [6]: Lasso regression is used to build the hypergraph, HCC is used as high-order features, and finally, the multi-kernel SVM [7] is used for classification.

UW-ElasticNet and UW-Lasso [8]: Elastic net and Lasso regression are used to build unweighted hypergraphs separately. Node degree, centrality degree, and efficiency are used as the classification features of SVM.

HL [9]: Sparse linear regression is used to construct the hypergraph, and the weights of the hypergraph are obtained by manifold learning. Finally, the similarity matrix is calculated according to the weights as the features for SVM classification.

sgLasso-COMHCC [10]: Spare group Lasso is used to build the hypergraph, HCC and COMHCC are used as high-order features, and finally, the multi-kernel SVM is used for classification.

Lasso-FCM [11]: FCM is constructed using lasso regression and dynamics of formula (3) in main paper, and FCMs are used as data features for SVM classification. Lasso-FCM can reconstruct large-scale gene regulatory networks due to its sparsity.

ElasticNet-HFCM [12]: High-order FCM (HFCM) is constructed using the elastic net and high-order dynamic formula [13], and the predicted values of HFCM are used as features for SVM classification. ElasticNet-HFCM is used to classify EEG data and has competitive performance.

Appendix C

Standard measures, including accuracy (ACC), sensitivity (SEN), specificity (SPE), F1 Score, and area under the receiver operating characteristic (ROC) curve (AUC), are used to evaluate the classification performance of different methods. The specific formula is defined as follows:

$$ACC = \frac{TP + TN}{TP + FN + TN + FP} \tag{1}$$

$$SEN = \frac{TP}{TP + FN} \tag{2}$$

$$SPE = \frac{TN}{TN + FP} \tag{3}$$

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

where

$$Precision = \frac{TP}{TP + FP} \tag{5}$$

$$Recall = SEN = \frac{TP}{TP + FN} \tag{6}$$

This work was supported in part by the Key Project of Science and Technology Innovation 2030 supported by the Ministry of Science and Technology of China under Grant 2018AAA0101302, in part by the National Natural Science Foundation of China under Grant 62206205, and in part by the Zhejiang Lab's International Talent Fund for Young Professionals.

Yingzhi Teng, Jing Liu, and Yifan Li are with the with the Guangzhou Institute of Technology, Xidian University, Guangzhou 510555, China (e-mail: ingeteng@163.com; neouma@mail.xidian.edu.cn; 18066899461@163.com).

Kai Wu (corresponding author) is with the School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: kwu@xidian.edu.cn).

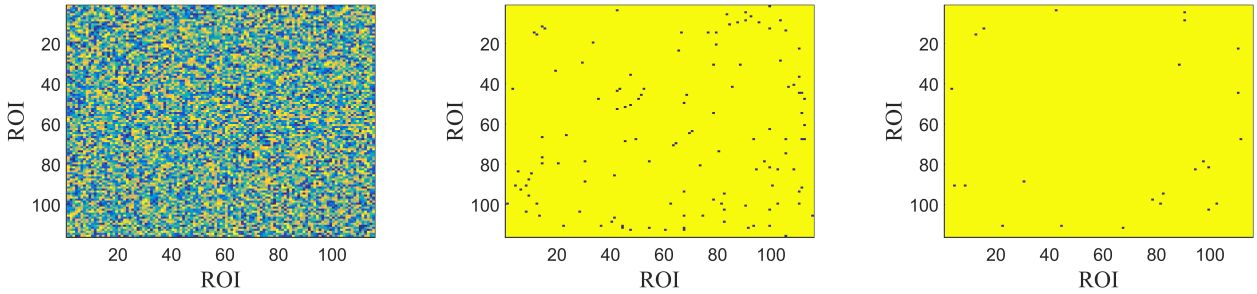
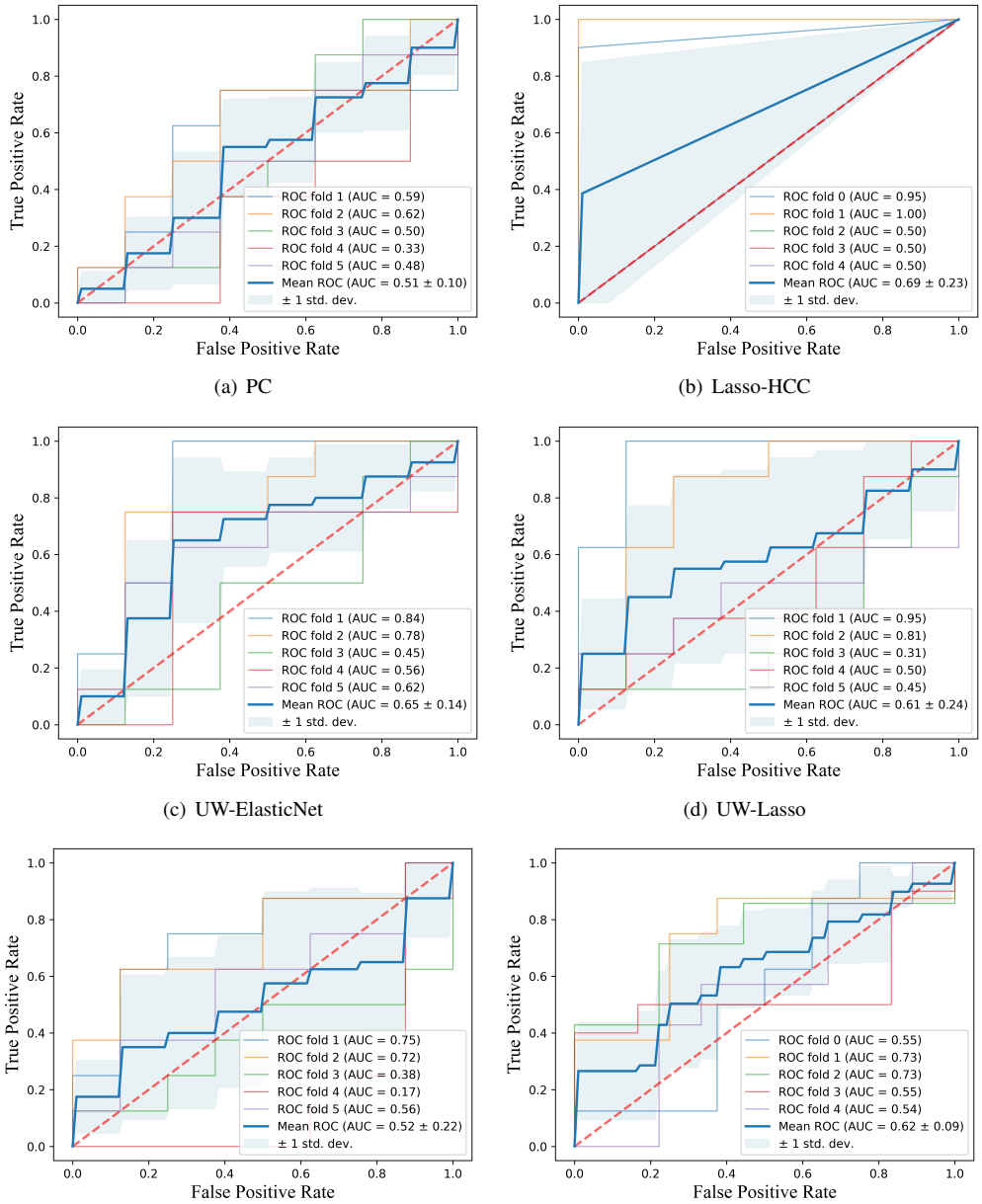
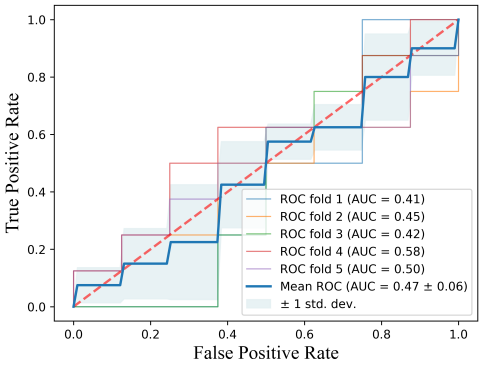
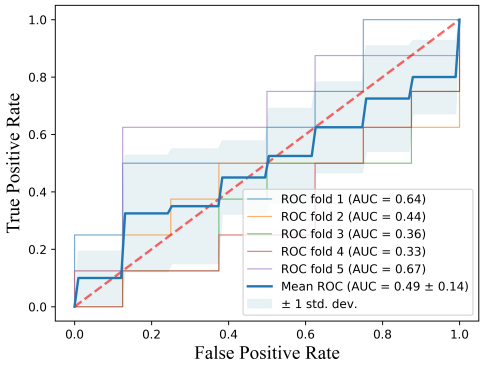


Fig. S1. Visualization of p-values on connections of FCNs between ROIs. (a) The p-values on connections between ROIs. (b) The connections with the p-values less than 0.005. (c) The connections with the p-values less than 0.001.

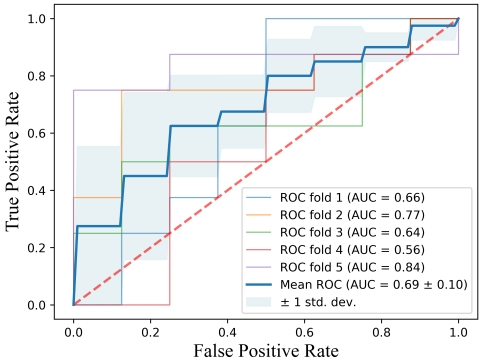




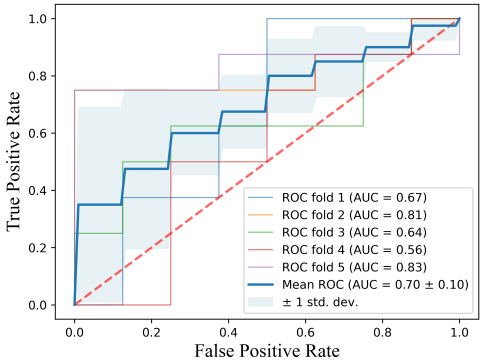
(g) Lasso-FCM



(h) ElasticNet-HFCM



(i) Hyper-FCM o/w hypergraph



(j) Hyper-FCM

Fig. S2. The 5-fold crossover ROC curves of methods including (a) PC, (b) Lasso-HCC, (c) UW-ElasticNet, (d) UW-Lasso, (e) HL, (f) sgLasso-COMHCC, (g) Lasso-FCM, (h) ElasticNet-HFCM, (i) Hyper-FCM o/w hypergraph, and (j) Hyper-FCM.

REFERENCES

[1] C. G. Yan, X. D. Wang, X. N. Zuo, and Y. F. Zang, "DPABI: Data processing & analysis for (resting-state) brain imaging," *Neuroinformatics,* vol. 14, no. 3, pp. 339–351, 2016.

[2] N. Tzourio-Mazoyer, B. Landeau, D. Papathanassiou, F. Crivello, O. Etard, N. Delcroix, B. Mazoyer, and M. Joliot, "Automated anatomical labeling of activations in SPM using a macroscopic anatomical parcellation of the MNI MRI single-subject brain," *NeuroImage,* vol. 15, no. 1, pp. 273–289, 2002.

[3] A. Baldassarre, C. M. Lewis, G. Committeri, A. Z. Snyder, G. Luca Romani, and M. Corbetta, "Individual variability in functional connectivity predicts performance of a perceptual task," *Proceedings of the National Academy of Sciences,* vol.109, no. 9, pp. 3516–3521, 2012.

[4] C.-Y. Wee, P.-T. Yap, D. Zhang, K. Denny, J. N. Browndyke, G. G. Potteret, K. A. Welsh-Bohmer, L. Wang, and D. Shen, "Identification of MCI individuals using structural and functional connectivity networks," *NeuroImage,* vol. 59, no. 3, pp. 2045–2056, 2012.

[5] R. Kashyap, R. Kong, S. Bhattacharjee, J. Li, J. Zhou, and B.-T. Thomas Yeo, "Individual-specific fMRI-Subspaces improve functional connectivity prediction of behavior," *NeuroImage,* vol. 189, pp. 804–812, 2019.

[6] B. Jie, C.-Y. Wee, D. Shen, and D. Zhang, "Hyper-connectivity of functional networks for brain disease diagnosis," *Medical Image Analysis,* vol. 32, pp. 84–100, 2016.

[7] F. R. Bach, G. R. Lanckriet, and M. I. Jordan, "Multiple kernel learning, conic duality, and the SMO algorithm," In: *Proceedings of the Twenty-first International Conference on Machine Learning,* 2004.

[8] H. Guo, Y. Li, Y. Xu, Y. Jin, J. Xiang, and J. Chen, "Resting-state brain functional hyper-network construction based on elastic net and group lasso methods," *Frontiers in Neuroinformatics,* vol. 12, no. 25, 2018.

[9] L. Xiao, J. Wang, P. H. Kassani, Y. Zhang, Y. Bai, J. M. Stephen, T. W. Wilson, V. D. Calhoun, and Y.-P. Wang,"Multi-hypergraph learning-based brain functional connectivity analysis in fMRI data," *IEEE Transactions on Medical Imaging,* vol. 39, no. 5, pp. 1746–1758, 2019.

[10] Y. Li, C. Sun, P. Li, Y. Zhao, G. K. Mensah, Y. Xu, H. Guo, and J. Chen, "Hypernetwork construction and feature fusion analysis based on sparse group lasso method on fMRI dataset," *Frontiers in Neuroscience,* vol. 14, no. 60, 2020.

[11] K. Wu and J. Liu, "Robust learning of large-scale fuzzy cognitive maps via the lasso from noisy time series," *Knowledge-Based Systems,* vol. 113, pp. 23–38, 2016.

[12] F. Shen, J. Liu, and K. Wu, "Multivariate time series forecasting based on elastic net and high-order fuzzy cognitive maps: a case study on human action prediction through EEG signals," *IEEE Transactions on Fuzzy Systems,* vol. 29, no. 8, pp. 2336–2348, 2020.

[13] S. Yang and J. Liu, "Time-series forecasting based on high-order fuzzy cognitive maps and wavelet transform," *IEEE Transactions on Fuzzy Systems,* vol. 26, no. 6, pp. 3391–3402, 2018.